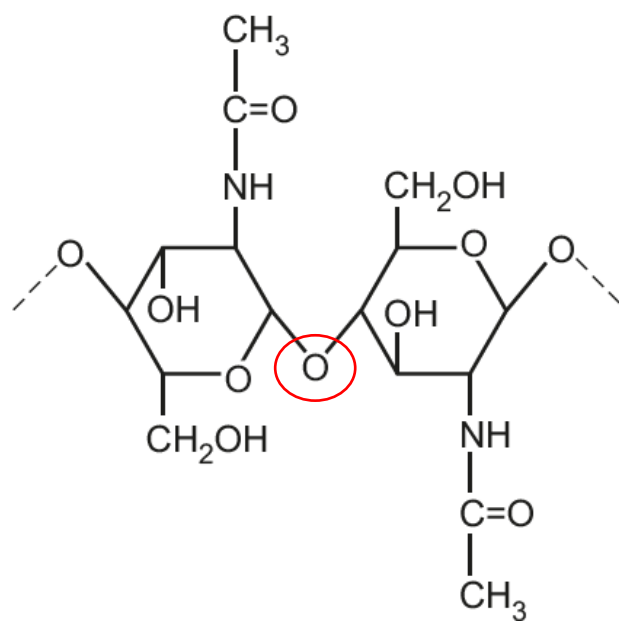
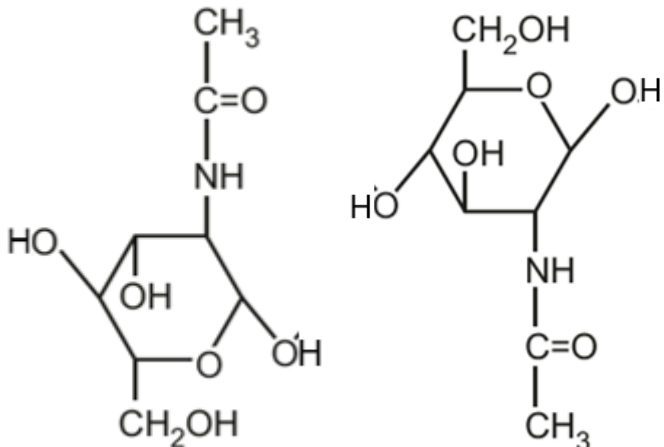


4(a)(i)



correct glycosidic bond circled ;

1

Question	Answer	Marks
4(a)(ii)	 -OH group shown on the monomer where the glycosidic bond connected the two monomers (C1) ; -OH shown on C4 ;	2
4(b)	<i>any four from:</i> <i>ref. to</i> filtering helps prevent / AW, cholera ; filtering water removed copepods (with bacteria on surface) from water ; some bacteria were able pass through fabric / bacteria were found living free in water / some copepods were small enough to pass through fabric ; cholera transmission could be from, other sources / food ; people who filter water may have better personal hygiene ; A examples e.g. washing hands AVP ; e.g. differences in other water treatment methods e.g. boiling water	4

Question	Answer	Marks
4(c)(i)	any <i>one</i> from: wash hands, qualified ; e.g. more frequently after, urination / defecation / using the toilet I non-scientific terms before preparing food before handling utensils thoroughly / with soap / antibacterial gel avoid eating, crops / food, grown with faeces used as fertiliser ; wash fruit / vegetable (in, uncontaminated / AW, water) ; cover food to prevent flies landing / kill flies before they land on food ; AVP ; e.g don't share hand towels use disposable hand towels keep fingernails short	1
4(c)(ii)	any <i>two</i> from: <i>ref. to active transport / ref. to efflux pumps ; in correct context</i> tetracycline transported out of the cell ; AVP ; e.g. substance pumped out combined with tetracycline (membrane) enzyme that breaks down tetracycline before entry to cell	2
4(c)(iii)	any <i>two</i> from: vaccines may contain more than one antigen (so more than one type of antibody is present) ; multiple mutations are needed to evolve, immune evasion / described; vaccines given to people before infection so secondary immunity acts quickly to kill pathogen before mutations arise ; antibiotics used when person already infected so large population ; ora more likely to be a mutation which causes resistance ; ora antibiotics may target only one site in a bacterial cell ; antibiotics more likely to be misused ; AVP ; ;	2